

Pedro José Campos de Almeida

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Portfolio: <https://pedroalmeid.github.io/portfolio>

PROFESSIONAL SUMMARY

Computer Science undergraduate student with solid technical experience, including Full-Stack development of digital products with a focus on innovation, and applied Data Science research, with an emphasis on Explainable Artificial Intelligence (XAI) and Causal Inference.

EDUCATION

Federal University of Juiz de Fora (UFJF)

Bachelor of Science in Computer Science

6th Semester

Expected Graduation: Jul/2028

PROFESSIONAL & ACADEMIC EXPERIENCE

CIMO Engenharia e Soluções Ambientais

Full-Stack Developer (Innovation Department)

Dec/2025 – Present

- Worked within an intrapreneurship model, as part of an autonomous student team responsible for the ideation and end-to-end development of technology solutions.
- Performed requirements gathering, specification, and analysis. Modeled business rules and database structures.
- Developed scalable backend systems in Java (Spring Boot) and microservices (Firebase Cloud Functions with Python), with a relational database (PostgreSQL).
- Integrated Artificial Intelligence through LLM consumption (Perplexity API).
- Developed front-end applications with SvelteKit.
- Developed UI designs using Figma.

Undergraduate Research in Data Science

CNPq Undergraduate Research (PIBIC Program)

Sep/2025 – Present

- Processed real-world agroclimatic data for training predictive models and conducting data analyses focused on carbon emissions mitigation.
- Applied Explainable Artificial Intelligence (XAI) techniques in practice, using SHAP, PDP Plots, and ICE for the auditing and validation of Machine Learning models.
- Performed causal and statistical modeling using graphs and Causal Inference analyses (DoWhy).
- Authored a systematic literature review published at the ICEIS 2026 international conference (International Conference on Enterprise Information Systems).

Undergraduate Research in AI + Software Engineering

UFJF Undergraduate Research (BIC Program)

Sep/2024 – Aug/2025

- Developed a Python (FastAPI) backend to automate the evaluation and selection of Classification (ML) models applied to IoT.
- Managed the model lifecycle (MLOps) using MLflow for experiment tracking.
- Integrated data pipelines and automatically trained models using PyCaret (AutoML).

PERSONAL PROJECTS

World Cup Causal Predictor

Causal Modeling and Data Engineering

May/2026 – Jul/2026

- Developed an end-to-end predictive simulator for the 2026 World Cup, isolating confounding factors through Causal Inference.

- Built an automated Python pipeline using time-weighted Poisson Regression, tactical clustering via K-Means, and a probabilistic engine corrected with the Dixon-Coles adjustment.
- Operated the model in real time (production) throughout the tournament, achieving **70.6% overall accuracy in predicting match outcomes (W/D/L)** across 102 simulated matches.
- Delivered elite statistical performance in the knockout stages (30 matches), achieving **86.7% outcome accuracy (W/D/L)**, **23.3% exact scoreline accuracy**, and a Mean Absolute Error (MAE) of just 0.58 goals for the winning team.

AWARDS & ACADEMIC PUBLICATIONS

- **International Conference Presentation (ICEIS 2026):** Lead author of a published review on the use of Explainable AI (XAI) for carbon emissions mitigation in agriculture.
- **CAPES University Talent Award (2024):** Recognition awarded by CAPES to the top 1,000 nationally ranked participants in a general knowledge examination.

COURSES & CERTIFICATIONS

- **XAI With Python (A Data Odyssey)** – Specialization in explainability algorithms and model auditing.
- **Machine Learning e Data Science com Python de A a Z** (Udemy) – Introduction to data science and machine learning with Python.
- **DevOps with Docker** (University of Helsinki) – Introduction to DevOps, software architecture, infrastructure, and containerization.

LEADERSHIP & VOLUNTEERING

- **Communications Coordinator | Projeto Buddy (ongoing):** Coordinator of the communications and design department of a voluntary academic outreach project focused on the internationalization of the Federal University of Juiz de Fora.
- **Committee Director | Model United Nations (MUN):** Led student-run debate events for over 200 students. Conducted training in public speaking, geopolitical negotiation, and conflict resolution, ensuring deliverables under pressure in a student-led environment.

TECHNICAL SKILLS

- **Software:** Python, Java, SQL, Svelte, RESTful Architecture, Microservices.
- **Data and Artificial Intelligence:** Machine Learning, DoWhy, XAI (SHAP), Pandas, Scikit-learn, MLOps.
- **Tools and Infrastructure:** Git, Docker, DevContainers, Figma, PostgreSQL, Firebase, FastAPI, MLflow.

LANGUAGES

- Portuguese (BR): Native;
- English: Advanced reading and comprehension; Intermediate writing and speaking